LESSON 2.10

Choose Appropriate Measures

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can choose the best measure of center for a data set based on its shape.

WHY THIS MATTERS The students want to describe the data they collected about the number of years of teaching experience of the teachers in their school. Which measure of center and measure of variation should they use to summarize the data set?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mean *Spanish: Media*	The average. Add all the numbers, then divide by how many there are.	Mean of 10, 20, 30 = (10+20+30) ÷ 3 = 20
Median *Spanish: Mediana*	The middle number when you put them in order.	Data: 5, 8, 12, 15, 20 → median is 12 (the 3rd of 5 values)
Outlier *Spanish: Valor atípico*	A number that is much bigger or smaller than the rest.	2, 3, 4, 32 → 32
Skewed *Spanish: Sesgado*	When most data sits on one side with a tail on the other.	A long tail to the right
Data distribution *Spanish: Distribución de datos*	How the data looks: where it sits and how spread out it is.	Symmetric = even on both sides. Skewed = bunched on one side with a tail
Variability *Spanish: Variabilidad*	How spread out the numbers are.	88, 90, 89, 91 (low variability) vs. 50, 70, 95, 100 (high variability)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Five friends saved $6, $7, $8, $9 and $50. Which center describes a typical friend?

1. 50 is an outlier.
2. Median = 8. Mean = 80 ÷ 5 = 16, more than four of the five values.
3. The median is closer to most of the values.

Answer: The median, $8

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Data set A: 4, 5, 6, 7, 8. Are there any outliers? If not, which measure should we use, mean or median?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The outlier in this set is _____ because _____ .
- The median is _____ and the mean is _____ .
- The mean is pulled _____ because _____ .
- I would report the _____ because _____ .

Word bank outlier | skewed | cluster | gap | median | mean | typical | center | pulled | symmetric | describe | data set

WATCH OUT!

A common mistake

A common mistake in Appropriate Measures is defaulting to the mean just because "it uses all the data," without first checking whether an outlier or skewed shape is present. For example, given a player's points per game 14, 16, 15, 17, 16, 58, a student calculates the mean (136 ÷ 6 ≈ 22.7) and reports that as the typical game

LESSON 2.10

Choose Appropriate Measures

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can choose the best measure of center for a data set based on its shape.

WHY THIS MATTERS The students want to describe the data they collected about the number of years of teaching experience of the teachers in their school. Which measure of center and measure of variation should they use to summarize the data set?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each data description with the best measure of center.

_____	Test scores: 88, 90, 85, 92, 87 (no outliers)	A	Mean
_____	Salaries: $30K, $32K, $35K, $31K, $250K	B	Median
_____	Race times: 12.1s, 12.3s, 12.0s, 12.2s, 12.1s	C	Mean
_____	Goals scored: 1, 2, 1, 0, 2, 1, 15	D	Median

HINT Scan each data list for a value that's wildly different from its neighbors — like one huge salary or one giant goal count.

2 **Write the name of the correct group on each line.**

Sort each feature by which measure of center it supports.

Groups: Use Mean | Use Median

_____ No outliers in the data

_____ Data is symmetric

_____ All values are close together

_____ Data has one or more outliers

_____ Data is skewed to one side

_____ One extreme value pulls the average

HINT The mean is sensitive to extreme values — outliers pull it up or down.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A gymnast's scores are: 8.5, 8.8, 8.7, 8.6, 8.9. There are no outliers. Which measure best represents a typical score?

(A) Median

(B) Neither — the data is too small

(C) Both are equally bad

(D) Mean

HINT Look at the scores: 8.5 up to 8.9 — all five are bunched tightly together with no unusual value.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A runner's mile times are: 7:10, 7:15, 7:12, 7:20, 12:00. The 12:00 was due to a cramp. Which measure best represents a typical mile?

(A) Mean — it uses all the data

(B) Median — the outlier 12:00 pulls the mean too high

(C) Mode — it shows the most common time

(D) Neither measure works

HINT Four times are near 7 minutes; one is 12:00 from a cramp. That 12:00 is an outlier, not a typical mile.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Data set: 5, 6, 7, 7, 8, 50. The mean is about 13.8 and the median is 7. Which better represents the typical value?

(A) Mean (13.8) — it includes all values

(B) Both are equally good

(C) Median (7) — the outlier 50 inflates the mean

(D) Neither works for this data

HINT Five of the six values are between 5 and 8 — then there's 50. Notice where the mean (13.8) lands compared to those five.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake — Forgetting to Order the Data

1. **Data** A swimmer's lap counts: 9, 12, 8, 11, 10 (no outliers).
2. **Find the median** The student keeps the numbers as written (9, 12, 8, 11, 10) and picks the middle one: median = 8.
3. **Choose a measure** The student reports the median (8) as the typical lap count.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT The median is the MIDDLE value only after the numbers are sorted from least to greatest.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you notice about when median is the better choice?

SENTENCE STARTER The median is better when the data has ___ because the mean gets pulled toward ___. The mean is better when the data is ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.10

Choose Appropriate Measures

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can choose the best measure of center for a data set based on its shape.

WHY THIS MATTERS The students want to describe the data they collected about the number of years of teaching experience of the teachers in their school. Which measure of center and measure of variation should they use to summarize the data set?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Data Set	Mean	Median	Outlier?	Best Measure
15, 18, 16, 17, 15				
15, 18, 16, 17, 72				

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match the reason with whether to use mean or median.

_____ Data is symmetric and has no extreme values

_____ Data has one or more outliers

_____ Data is skewed to one side

_____ All values are clustered close together

(A) Use Mean

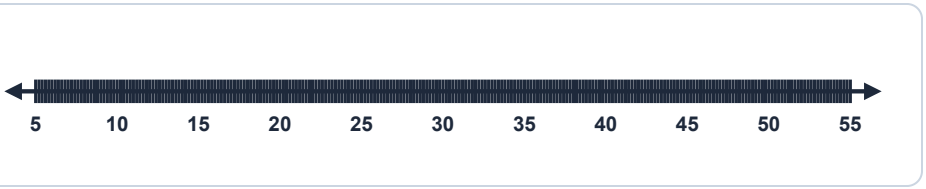
(B) Use Median

(C) Use Median

(D) Use Mean

3 **Mark and label each value on the number line.**

Data: 10, 12, 11, 13, 50. Place the mean (19.2) and median (12) on the number line. Notice how far apart they are because of the outlier.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A baseball player's batting averages over 6 seasons are: .280, .295, .290, .285, .300, .110. The .110 was an injury-shortened season. Which measure better represents the player's typical batting average?

(A) Median, because the .110 outlier pulls the mean down

(B) Mean, because it uses all the data

(C) Mean, because .110 is a real season

(D) Neither measure works

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Mistake — Choosing the Mean Despite an Outlier

1. **Data** A receiver's catches per game: 12, 14, 13, 15, 11, 60 (the 60 was a record-setting playoff game).
2. **Find the mean** Sum = 125, Count = 6, Mean = 125 ÷ 6 ≈ 20.8.
3. **Find the median** Ordered: 11, 12, 13, 14, 15, 60 → Median = (13 + 14) ÷ 2 = 13.5.
4. **Choose a measure** The student reports the mean (≈ 20.8) as the typical number of catches because it uses every game.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you notice about when median is the better choice?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.10

Choose Appropriate Measures

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can choose the best measure of center for a data set based on its shape.

WHY THIS MATTERS The students want to describe the data they collected about the number of years of teaching experience of the teachers in their school. Which measure of center and measure of variation should they use to summarize the data set?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort by which measure is typically used in real life.

Groups: Mean is typically reported | Median is typically reported

_____ Average temperature for the month

_____ Average test score for a class (no outliers)

_____ Median household income in a city

_____ Median home price in a neighborhood

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the Reasoning Error

1. **Data** Player minutes per game: 32, 34, 30, 35, 33, 5
2. **Find mean** Sum = 169, Count = 6, Mean = 28.2
3. **Find median** Ordered: 5, 30, 32, 33, 34, 35 → Median = (32+33) ÷ 2 = 32.5
4. **Conclusion** The mean (28.2) is the best measure because it uses all the data.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

A local newspaper reports that the average home price in a neighborhood is $450,000. The actual prices are: $180K, $200K, $190K, $210K, $195K, $1,725K. Is the reporter's claim misleading? Calculate the mean and median and explain which better represents a 'typical' home price.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What pattern do you notice about when median is the better choice?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help